

ray_limit: selectivity ray, complete-reaction limit, and closure profile $C(f)$

Mixing-limited reaction closure — `species_limits.py, integrate_species_odes`

Setup

Active species $i \in \mathcal{A}$, reactions $j = 1, \dots, R$. Net stoichiometry matrix

$$N \in \mathbb{R}^{|\mathcal{A}| \times R}, \quad N_{ij} = \nu_{ij}^{\text{prod}} - \nu_{ij}^{\text{reac}}.$$

Feed vectors $Y^{(1)}, Y^{(2)} \in \mathbb{R}^{|\mathcal{A}|}$ (stream 1 at $f = 1$, stream 2 at $f = 0$). The **no-reaction line** at mixture fraction $f \in [0, 1]$ is

$$M_i(f) = f Y_i^{(1)} + (1 - f) Y_i^{(2)}.$$

Let y be the current ODE state, y^0 its initial value, and $\langle \cdot \rangle$ the expectation under the step's $\text{Beta}(\alpha_t, \beta_t)$ pdf.

1. Selectivity ray

The accumulated change $\Delta := y - y^0$ lies in $\text{range}(N)$. The **selectivity ray** is taken directly as this accumulated extent:

$$d = \Delta = y - y^0.$$

This preserves the true reaction direction without distortion from a nonnegativity clip — important when products form by more than one route (parallel reactions, intermediates). Near y^0 (where $\Delta \approx 0$ and the direction is ill-conditioned) the code falls back to the mass-action projection $d = N r(y)$, where $r(y)$ is the vector of instantaneous mass-action rates.

The ray d encodes the **current selectivity**: which species have been consumed or produced and in what ratio. It rotates as the ODE state evolves, tracking the shift between competing pathways.

2. The complete-reaction limit $B(f)$

$B_i(f)$ is built by reacting the mixing line $M(f)$ to completion along the ray d — advancing as far as possible without any species going negative.

Maximum extent. At each f the largest non-negative scalar $c_{\max}(f)$ that keeps $M(f) + c d \geq 0$ is

$$c_{\max}(f) = \min_{i: d_i < 0} \frac{M_i(f)}{-d_i}.$$

Only consumed species ($d_i < 0$) constrain $c_{\max}$; products ($d_i > 0$) are unconstrained. The limit is then

$$B_i(f) = M_i(f) + c_{\max}(f) d_i.$$

3. Closure profile $C_i(f)$ and scalar λ

The closure profile interpolates between the complete-reaction limit B_i and the no-reaction line M_i :

$$C_i(f) = B_i(f) + [M_i(f) - B_i(f)] \lambda_i$$

$\lambda_i = 0$ gives the pure complete-reaction limit; $\lambda_i = 1$ gives pure no-reaction mixing. λ_i is set so that $\langle C_i \rangle = y_i$:

$$\lambda_i = \text{clip}_{[0,1]} \frac{y_i - \langle B_i \rangle}{\langle M_i \rangle - \langle B_i \rangle}.$$

Segment integrals use the regularised incomplete Beta function $I_x(\alpha, \beta)$ (betainc):

$$\int_{f_a}^{f_b} (sf + b) p_\beta df = s \frac{\alpha_t}{\alpha_t + \beta_t} [I_{f_b}(\alpha_t + 1, \beta_t) - I_{f_a}(\alpha_t + 1, \beta_t)] + b [I_{f_b}(\alpha_t, \beta_t) - I_{f_a}(\alpha_t, \beta_t)].$$

Global λ for pure cross-stream. Since M_i is linear, $\langle M_i \rangle = M_i(\bar{f}) = y_i^0$, and $B_i - M_i = c_{\max} d_i$ by construction, so

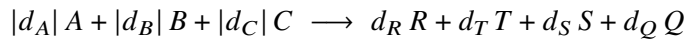
$$\lambda_i = \frac{y_i - \langle B_i \rangle}{y_i^0 - \langle B_i \rangle} = \frac{d_i(1 - \langle c_{\max} \rangle)}{-\langle c_{\max} \rangle d_i} = 1 - \frac{1}{\langle c_{\max} \rangle}$$

— **the same scalar for every species** — and $\langle C_i \rangle = y_i$ exactly (no clamp needed). Per-species λ_i diverge from this only when the clamp $[0, 1]$ binds.

4. Worked illustration: input_fuller_BC_azo_coupling_kinetics

Scheme: R1: $A+B \rightarrow R$, R2: $A+R \rightarrow S$, R3: $A+B \rightarrow T$, R4: $A+T \rightarrow S$, R5: $A+C \rightarrow Q$. **Feeds:** $Y^{(1)} = 15 A$ (stream 1), $Y^{(2)} = 1.2 B + 1.2 C$ (stream 2); $f_{\text{mean}} = 0.0625$.

The effective single reaction. The selectivity ray $d = \Delta$ records the net moles of each species consumed or produced per unit of accumulated reaction progress. Its components d_i are the net stoichiometric coefficients of a single *effective lumped reaction* that encodes the current mixture of all five pathways:

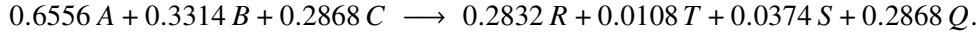


The coefficients rotate as the ODE state evolves. The table gives them at each trajectory's output-grid time-midpoint ($t = t_{\text{end}}/2$) for three dissipation rates ε — the actual A conversion reached at that instant is 77%, 70%, and 92% respectively, not a fixed 50%; the extent of reaction still differs substantially at the same *relative* point because the trajectories are far from self-similar across ε . At low ε (broad pdf, persistent A -rich pockets) the secondary pathways dominate and S carries most of the product weight; at high ε (near-uniform, A -lean) the primary reactions dominate and R and Q accumulate.

	$\varepsilon = 10^{-6}$	$\varepsilon = 1$	$\varepsilon = 10^6$
(α_i, β_i)	(0.311, 4.66)	(0.372, 5.58)	(14.5, 218)
kink f_s^*	0.1927	0.1366	0.0781
λ (global)	0.000 [†]	0.0008	0.0541
d_A	-0.7225	-0.6556	-0.8616
d_B	-0.2422	-0.3314	-0.8135
d_C	-0.2421	-0.2868	-0.0470
d_R	+0.0040	+0.2832	+0.7560
d_T	+0.0000	+0.0108	+0.0564
d_S	+0.2382	+0.0374	+0.0011
d_Q	+0.2421	+0.2868	+0.0470

[†] At $\varepsilon = 10^{-6}$ the mean has reached the complete-reaction average ($y_i \geq \langle B_i \rangle$), so λ clamps at 0 and $\langle C_i \rangle = \langle B_i \rangle$.

Building the limit ($\varepsilon = 1$). The effective reaction at this step is



The complete-reaction limit $B(f)$ is found by advancing the mixing line $M(f)$ along this reaction as far as possible without a reactant going negative. The maximum extent $c_{\max}(f)$ is set by whichever reactant is first exhausted:

$$c_{\max}(f) = \min\left(\frac{M_A(f)}{0.6556}, \frac{M_B(f)}{0.3314}, \frac{M_C(f)}{0.2868}\right) = \min\left(\frac{15f}{0.6556}, \frac{1.2(1-f)}{0.3314}, \frac{1.2(1-f)}{0.2868}\right).$$

Since $1.2(1-f)/0.3314 < 1.2(1-f)/0.2868$ for all f , the binding constraints are A and B . Using $p_i = M_i(0)/|d_i|$, $q_i = (M_i(1) - M_i(0))/|d_i|$:

$$(p, q)_A = (0, 22.88), \quad (p, q)_B = (3.621, -3.621), \quad (p, q)_C = (4.183, -4.183).$$

The A - and B -lines cross at the kink: $f_s^* = 3.621/26.50 = 0.1366$, $c^* = 3.127$. The peak values $v_i^{\text{fs}} = M_i(f_s^*) + c^* d_i$ (e.g. $v_R^{\text{fs}} = 0 + 3.127 \times 0.2832 = 0.886$) complete B_i ; their Beta-averages give $\langle B_i \rangle$, and $C_i = B_i + (M_i - B_i)\lambda$ with $\lambda = 0.0008$:

$\varepsilon = 1$	$\langle M_i \rangle = y_i^0$	$v_i^{\text{fs}} \text{ (mol m}^{-3}\text{)}$	$\langle B_i \rangle \text{ (mol m}^{-3}\text{)}$	$y_i \text{ (mol m}^{-3}\text{)}$	λ_i	$\langle C_i \rangle$
A	0.9375	0.000	0.2814	0.2819	0.0008	0.2819
B	1.1250	0.000	0.7934	0.7936	0.0008	0.7936
C	1.1250	0.139	0.8379	0.8382	0.0008	0.8382
R	0.0000	0.886	0.2835	0.2832	0.0008	0.2832
T	0.0000	0.034	0.0108	0.0108	0.0008	0.0108
S	0.0000	0.117	0.0374	0.0374	0.0008	0.0374
Q	0.0000	0.897	0.2871	0.2868	0.0008	0.2868

Every λ_i equals the global scalar and $\langle C_i \rangle = y_i$ to the quoted digits.

Note: single interior kink

Single interior kink. For a pure cross-stream scheme the consumed species are absent from their respective feed endpoints, so $c_{\max} = 0$ at $f = 0$ and $f = 1$. Between the endpoints $c_{\max}(f)$ is piecewise linear and

concave; its kink $f_s^\star$ is where the two tightest consumed-species constraints are simultaneously binding:

$$f_s^\star = \frac{p_b - p_a}{q_a - q_b}, \quad p_i = \frac{M_i(0)}{-d_i}, \quad q_i = \frac{M_i(1) - M_i(0)}{-d_i}.$$

The peak extent at the kink is $c^\star = c_{\max}(f_s^\star)$. This gives the **two-segment tent**:

$$B_i(f) = \begin{cases} M_i(f) + \frac{f}{f_s^\star} c^\star d_i & f \leq f_s^\star \\ M_i(f) + \frac{1-f}{1-f_s^\star} c^\star d_i & f > f_s^\star \end{cases}$$

with control-point peak value $v_i^{\text{fs}} = M_i(f_s^\star) + c^\star d_i$ and pure-feed endpoints $B_i(0) = Y_i^{(2)}$, $B_i(1) = Y_i^{(1)}$.

This structure holds when every species with $d_i < 0$ satisfies $Y_i^{(1)} = 0$ or $Y_i^{(2)} = 0$.

Intermediates are safe. A species absent from both feeds has $y_i^0 = 0$, and since concentrations are non-negative, $d_i = y_i - y_i^0 = y_i \geq 0$. So an intermediate can never have $d_i < 0$; the condition is vacuously satisfied for it.

Condition failure does not cause collapse. If a feed species appears in both streams ($Y_i^{(1)} > 0$ and $Y_i^{(2)} > 0$) and $d_i < 0$, then $M_i(f) > 0$ for all f and $c_{\max}$ remains positive. The only consequence is that $B(f)$ loses its endpoint-pinning ($B(0) \neq Y^{(2)}$ or $B(1) \neq Y^{(1)}$); the piecewise-linear machinery still computes the limit correctly.